

Solution Requirements

Date	15 March 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Bank analysts securely log in using username and password before accessing the prediction system</i>	High
2	Authorization levels	<i>Different user roles such as Admin, Credit Analyst, and Compliance Officer have role-based access permissions.]</i>	High
3	External interfaces	<i>Integrates with IBM Watson Machine Learning deployment, Flask web application, and applicant database for predictions.</i>	High

4	Transactions processing	<i>The system receives applicant details, preprocesses data, predicts approval status, and stores prediction results.]</i>	High
5	Reporting	<i>Generates prediction reports showing approval/rejection status, applicant details, and model performance statistics.</i>	Medium
6	Business rules, etc.	<i>Applicants with poor credit history, low income, or high loan balances are classified as high-risk according to predefined rules.]</i>	High
7	Compliance to laws or regulations	<i>Applicant data is processed securely while following banking data privacy and security guidelines.</i>	High
8	Other	<i>Provides real-time credit card eligibility prediction through a user-friendly web interface.</i>	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>The system should generate prediction results quickly for every application.</i>	<i>Prediction response within 2 seconds</i>

2	Scalability	<i>The system should support multiple users and increasing application requests without performance degradation.</i>	<i>Supports 500+ concurrent users</i>
3	Security & Data Privacy	<i>The system should support multiple users and increasing application requests without performance degradation.</i>	<i>Supports 500+ concurrent users</i>
4	Reliability & Availability	<i>The application should remain available with minimal downtime.</i>	<i>99.9% uptime</i>
5	Usability & Accessibility	<i>The web interface should be simple, responsive, and easy for analysts and customers to use.</i>	<i>Responsive design with intuitive navigation</i>
6	Other	<i>The solution should be maintainable, portable, and allow future model updates without affecting the application.</i>	<i>Supports easy deployment and model replacement</i>